

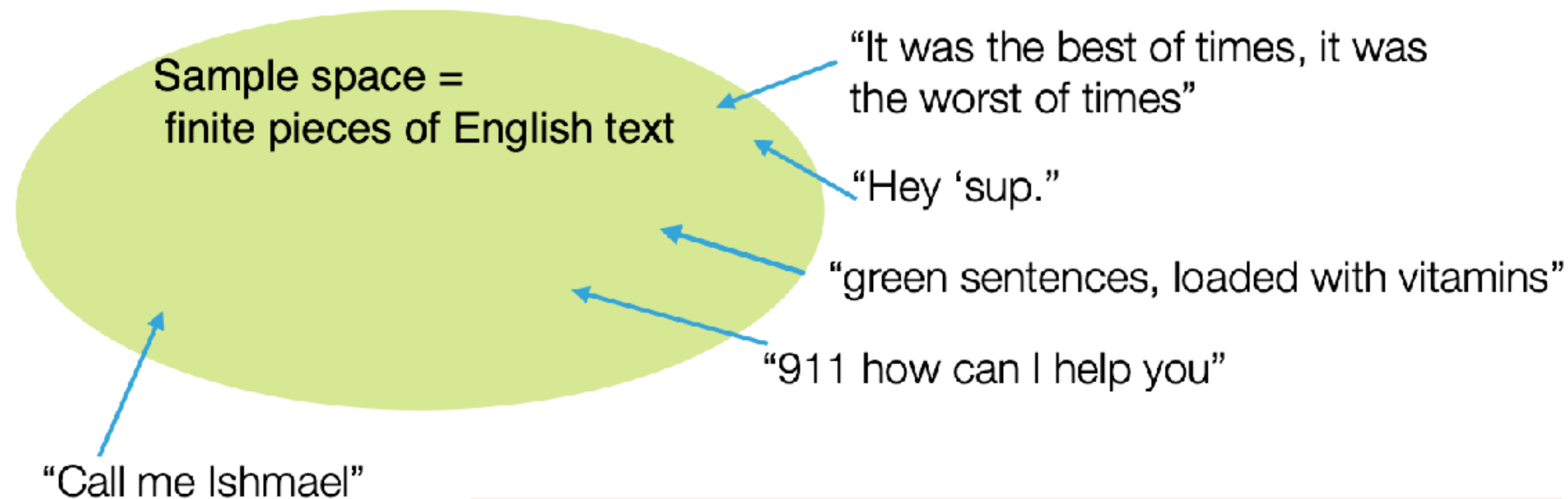
CPSC 8440: Generative AI: Technical and Human Perspectives

Module 1.4: Traditional Language Models

What is a language model?

- A probabilistic model of a sequence of words
- Joint probability distribution of words $w_1, w_2, \dots, w_n$:

$$P(w_1, w_2, w_3, \dots, w_n)$$



How likely is a given phrase, sentence, paragraph or even a document?

(i.e., $\Pr[w_1 w_2 w_2 \dots w_n]$ associated with every finite word sequence $w_1 w_2 w_2 \dots w_n$ (including nonsensical ones)

Chain rule

Conditional probability:
 $p(w \mid w_1, w_2), \forall w \in V$

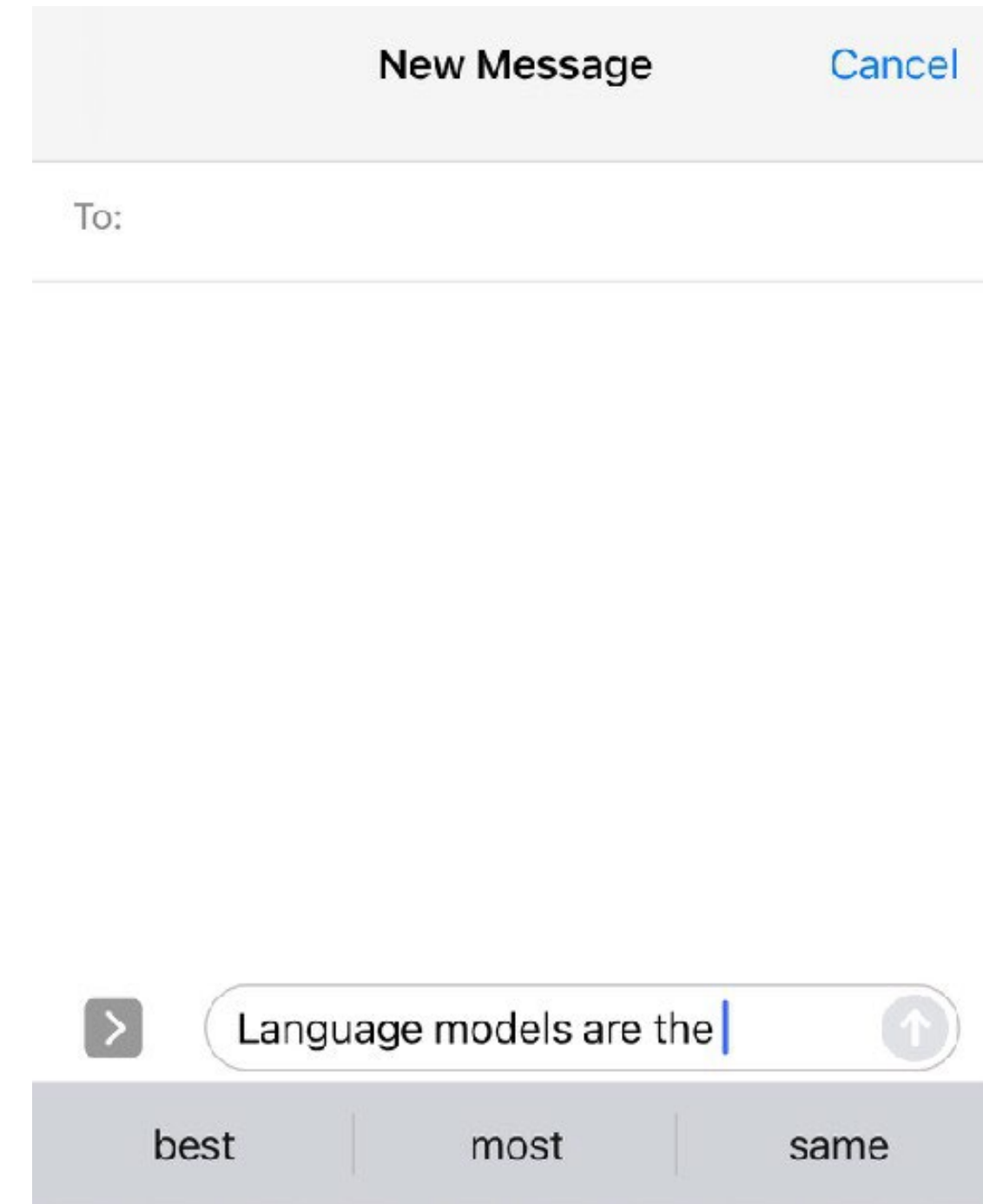
$$p(w_1, w_2, w_3, \dots, w_n) = p(w_1)p(w_2 \mid w_1)p(w_3 \mid w_1, w_2) \times \dots \times p(w_n \mid w_1, w_2, \dots, w_{n-1})$$

Sentence: “the cat sat on the mat”

$$\begin{aligned} P(\text{the cat sat on the mat}) = & P(\text{the}) \times P(\text{cat} \mid \text{the}) \times P(\text{sat} \mid \text{the cat}) \\ & \times P(\text{on} \mid \text{the cat sat}) \times P(\text{the} \mid \text{the cat sat on}) \\ & \times P(\text{mat} \mid \text{the cat sat on the}) \end{aligned}$$

Implicit order

Language models are everywhere



Estimating probabilities



$$P(\text{sat}|\text{the cat}) = \frac{\text{count}(\text{the cat sat})}{\text{count}(\text{the cat})}$$

$$P(\text{on}|\text{the cat sat}) = \frac{\text{count}(\text{the cat sat on})}{\text{count}(\text{the cat sat})}$$

⋮

Assume we have a vocabulary of size V ,
how many sequences of length n do we have?

A) $n * V$

B) n^V

C) V^n

D) V/n

Estimating probabilities



$$P(\text{sat}|\text{the cat}) = \frac{\text{count}(\text{the cat sat})}{\text{count}(\text{the cat})}$$

$$P(\text{on}|\text{the cat sat}) = \frac{\text{count}(\text{the cat sat on})}{\text{count}(\text{the cat sat})}$$

⋮

- With a vocabulary of size V , # sequences of length $n = V^n$
- Typical English vocabulary $\sim 40\text{k}$ words
- Even sentences of length ≤ 11 results in more than 4×10^{50} sequences.
Too many to count!
- (*For reference, # of atoms in the earth $\sim 10^{50}$*)

Markov assumption

- Use only the recent past to predict the next word
- Reduces the number of estimated parameters in exchange for modeling capacity
- 1st order

$$P(\text{mat} | \text{the cat sat on the}) \sim P(\text{mat} | \text{the})$$

- 2nd order

$$P(\text{mat} | \text{the cat sat on the}) \sim P(\text{mat} | \text{on the})$$



Andrey Markov

n-gram models

Unigram $P(w_1, w_2, \dots w_n) = \prod_{i=1}^n P(w_i)$ e.g. $P(\text{the}) P(\text{cat}) P(\text{sat})$

Bigram $P(w_1, w_2, \dots w_n) = \prod_{i=1}^n P(w_i | w_{i-1})$ e.g. $P(\text{the}) P(\text{cat} | \text{the}) P(\text{sat} | \text{cat})$

and Trigram, 4-gram, and so on.

*Larger the n , more accurate and better the language model
(but also higher costs)*

Estimating probabilities



Consider the following corpus

<s> I like apples </s>

<s> You like strawberries </s>

<s> You like apples </s>

Note: <s> and </s> are
starting and ending tokens

What's the bigram probability $P(\text{apples} \mid \text{like})$?

(A) $1/3$

(B) $2/3$

(C) $1/2$

(D) 1

Estimating probabilities



Consider the following corpus

<s> I like apples </s>

<s> You like strawberries </s>

<s> You like apples </s>

Note: <s> and </s> are
starting and ending tokens

What's the bigram probability $P(\text{apples} \mid \text{like})$?

(A) $1/3$

(B) $2/3$

(C) $1/2$

(D) 1

$$P(\text{apples} \mid \text{like}) = \frac{\text{Count}(\text{"like apples"})}{\text{Count}(\text{"like"})} = \frac{2}{3}$$

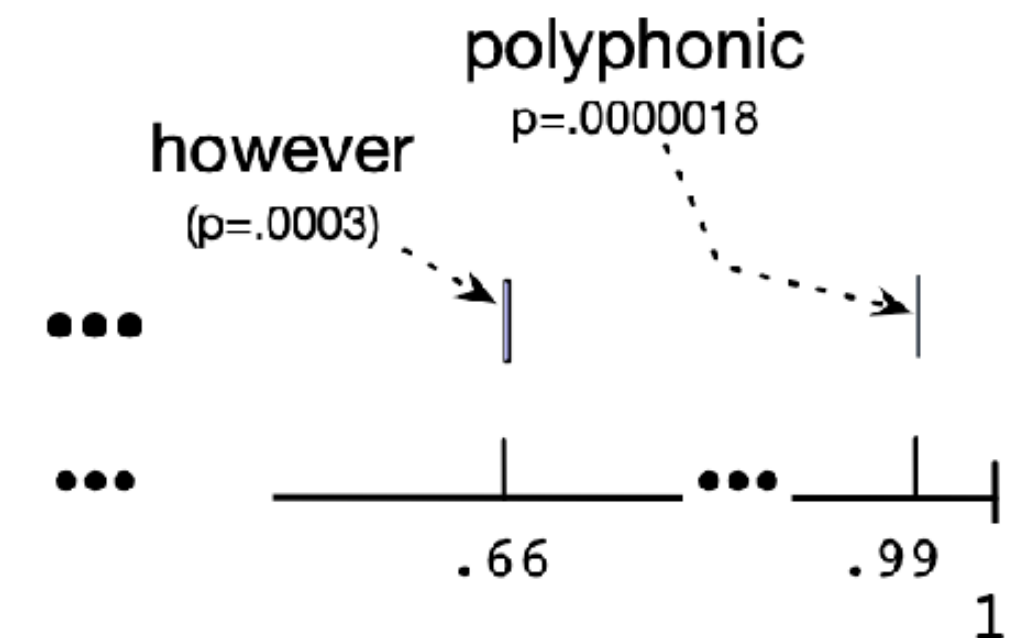
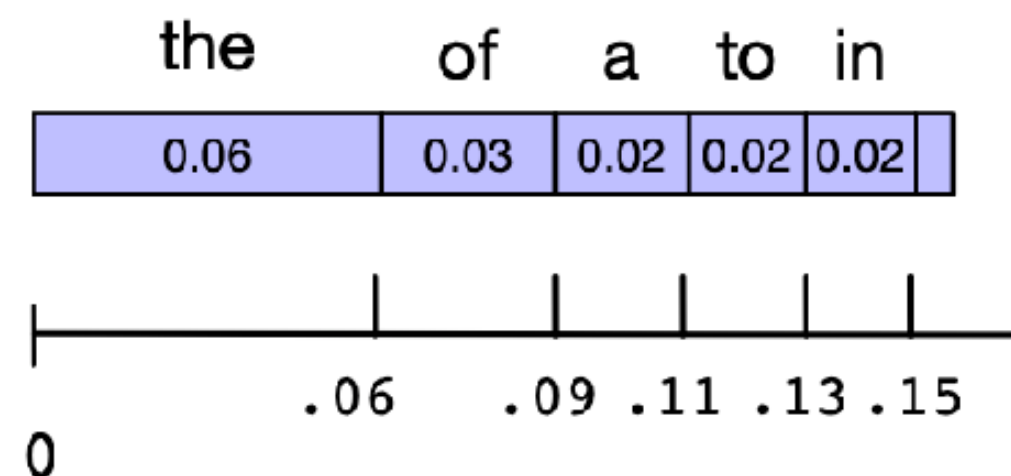
Generating from a language model

Generating from a language model

- Given a language model, how to generate a sequence?

Bigram
$$P(w_1, w_2, \dots, w_n) = \prod_{i=1}^n P(w_i | w_{i-1})$$

- Generate the first word $w_1 \sim P(w)$
- Generate the second word $w_2 \sim P(w | w_1)$
- Generate the third word $w_3 \sim P(w | w_2)$
- ...



Generating from a language model

- Given a language model, how to generate a sequence?

Trigram
$$P(w_1, w_2, \dots, w_n) = \prod_{i=1}^n P(w_i \mid w_{i-2}, w_{i-1})$$

- Generate the first word $w_1 \sim P(w)$
- Generate the second word $w_2 \sim P(w \mid w_1)$
- Generate the third word $w_3 \sim P(w \mid w_1, w_2)$
- Generate the fourth word $w_4 \sim P(w \mid w_2, w_3)$
- ...

Intrinsic evaluation of language models

- **Train** parameters on a suitable training corpus
 - Assumption: observed sentences $\sim$ good sentences
- **Test** on different, unseen corpus
 - If a language model assigns a higher probability to the test set, it is better
- Evaluation metric - **perplexity!**



Perplexity (ppl)

- Measure of how well a LM **predicts** the next word
- For a test corpus with words $w_1, w_2, \dots, w_n$

$$\text{Perplexity} = P(w_1, w_2, \dots, w_n)^{-1/n}$$

$$\text{ppl}(S) = 2^x \quad \text{where} \quad x = -\frac{1}{n} \log_2 P(w_1, \dots, w_n) = -\frac{1}{n} \sum_{i=1}^n \log_2 P(w_i | w_1 \dots w_{i-1})$$

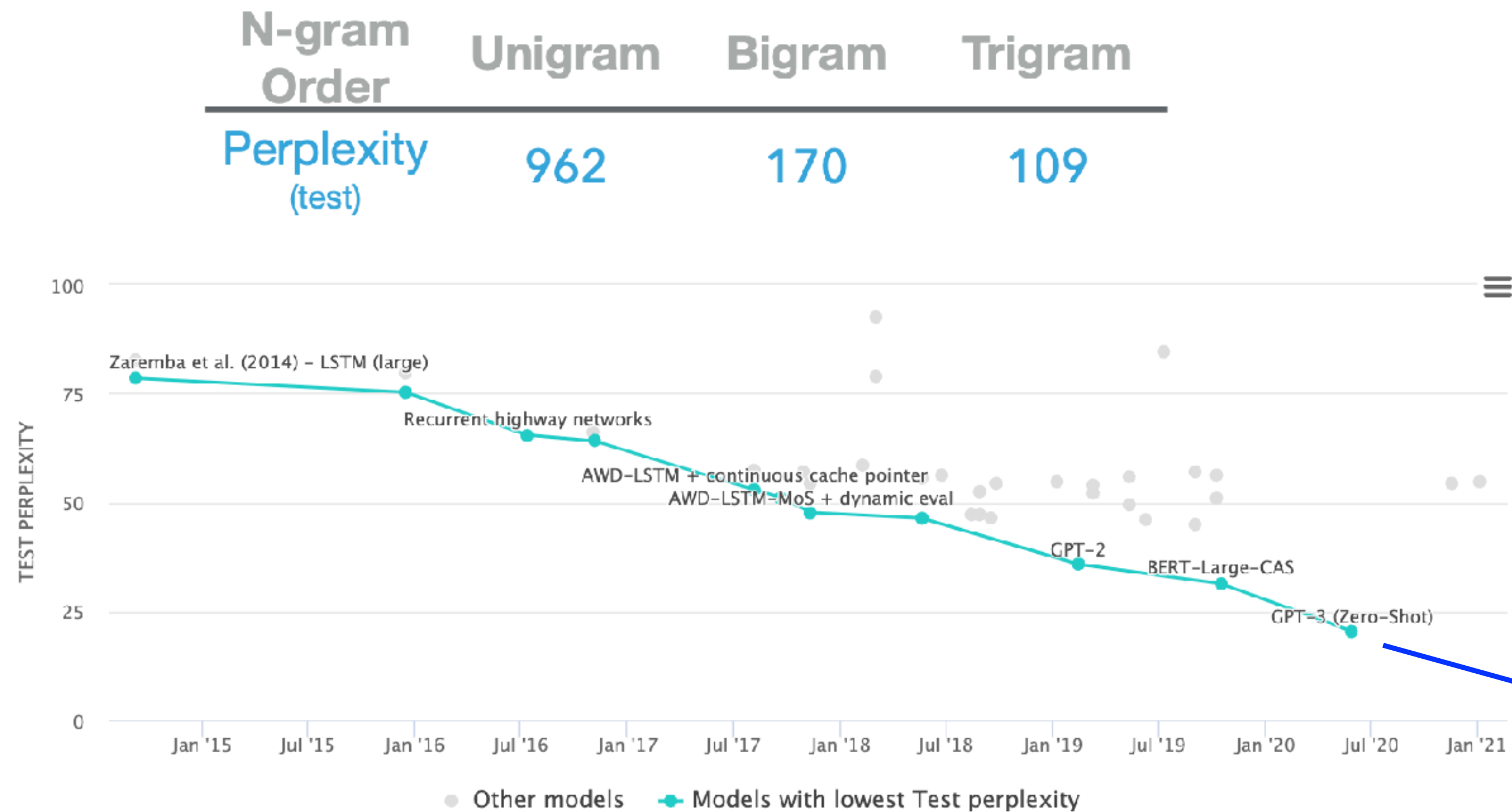
Cross-Entropy



- Minimizing perplexity $\sim$ maximizing probability of corpus

Perplexity

Training corpus 38 million words, test corpus 1.5 million words, both **WSJ**



GPT-3 175B:
ppl = 20.5

<https://paperswithcode.com/sota/language-modelling-on-penn-treebank-word>